

SMART: A Selective Marker-guided Adaptive Recursive Transformer for Genomic Classification

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Abstract

Transformer models for genomic classification treat every one of the thousands of measured genes as an equally important token and stack many independent layers, making them parameter-heavy and leaving *which* genes deserve computation to be learned implicitly. We argue that for genomics, parameter efficiency should be an *architectural* property and computation should be *allocated adaptively*. We present **SMART**, which (i) compresses thousands of genes into a small set of interpretable tokens – learned marker genes for single-cell expression, or curated Reactome *pathway* tokens for multi-omics – cutting self-attention from $\mathcal{O}(N^2)$ to $\mathcal{O}(M^2)$; (ii) processes those tokens with a *single* transformer block applied recursively, so parameters do not grow with depth; and (iii) uses a Mixture-of-RecurSIONs router to give each token its own recursion depth, turning depth into an intrinsic compute-allocation importance score. We validate the three resulting claims – **token reduction**, an **adaptive recursion loop**, and **parameter reduction** – with a controlled study across the genomap single-cell datasets and Reactome/P-NET multi-omics cohorts, with no TCGA bulk data. Weight sharing yields an exact $K \times$ parameter reduction (a $4 \times$ reduction at $K=4$) at accuracy comparable to independent layers; adaptive depth leaves a mean depth well below K and saves roughly a third of the recursion compute at parity with fixed depth; and a few dozen to a few hundred interpretable tokens recover most of the full-gene accuracy. We report negative results transparently: adaptive routing helps only when the task has computational slack, and label-free biological priors give small gains. The whole pipeline – training, ablations, and this paper – regenerates from one command.

1 Introduction

Single-cell RNA sequencing and tumour multi-omics now profile thousands of genes per sample, and transformer models such as scGPT, Geneformer and scBERT have adapted the architecture of Vaswani et al. (2017) to this modality. They inherit two costly habits from language models. First, they treat *every* gene as an equally important token, so a

housekeeping gene and a lineage-defining marker receive the same budget and attention scales quadratically in the number of genes. Second, they stack *independent* layers, so parameters grow linearly with depth.

SMART rejects both. It keeps only a small set of *interpretable tokens* and applies a *single* weight-shared transformer block recursively, with a router that gives each token its own recursion depth. This is, in spirit, a biological instantiation of Mixture-of-RecurSIONs (Bae et al., 2025): a recursive, adaptively-routed transformer. Because SMART is a one-shot *set* classifier rather than an autoregressive model, it is not obvious which of those design decisions still pay off. We therefore run a controlled study of all of them and organise the paper around the three claims they support.

Contributions. (1) SMART, a recursive transformer that makes parameter efficiency architectural and computation adaptive, with interpretable marker- or pathway-token inputs. (2) A controlled validation of three claims – token reduction, adaptive recursion, parameter reduction – on the genomap single-cell suite and Reactome/P-NET multi-omics, with no TCGA. (3) Transparent negative results: where adaptive routing and biological priors do and do not help.

2 The SMART Model

Interpretable tokens. A gene-identity embedding plus a value projection embeds each input feature. SMART then forms $M \ll N$ tokens. For single-cell expression these are learned marker slots produced by a cross-attention router whose queries attend over all genes; for multi-omics they are the curated Reactome pathways, each pooling its member genes’ mutation, copy-number and expression channels (mean pooling for dense assays, burden pooling for sparse mutation). Either way the tokens carry biological meaning and the downstream attention costs $\mathcal{O}(M^2)$.

Recursive stack and parameter sharing. One pre-norm transformer block is applied for K recursion steps. Sharing across the K steps ranges from a single block (full sharing) to K independent blocks, with graded Cycle/Sequence/Middle-Cycle/Middle-Sequence schemes in between.

Adaptive recursion depth. A Mixture-of-R recursions router allocates depth per token. Expert-choice routing keeps a capacity top- k of tokens at each step and funnels survivors deeper, so a token’s survival depth is an importance score; token-choice routing lets each token pick one depth. A label-free biological prior (co-expression centrality, or the Reactome pathway-hierarchy graph) can bias the depth logits.

3 Experimental Setup

Data. *Genomap single-cell suite* (genomap features; Islam and Xing, 2023); *Tabula Muris*, pancreas, common.class and prototype from the genomap capsule, plus Baron and Segerstolpe rebuilt from raw counts via genomap construction. *Reactome/P-NET multi-omics* (Elmarakeby et al., 2021): prostate, bladder, stomach and breast cohorts, and pan-cancer metastatic-vs-primary and 32-class cancer-type tasks. No TCGA bulk data is used. We follow the genomap-paper protocol and report macro-F1 (mean $\pm$ std over seeds where available).

4 Results

We organise results by claim. **Token reduction:** accuracy holds as the number of interpretable tokens M is reduced (Table 1), and learning which genes are markers is competitive with or better than fixed panels (Table 2). **Adaptive recursion:** adaptive per-token depth matches fixed depth while leaving the mean depth well below K and saving roughly a third of the recursion compute (Table 3, Fig. 2); the gain is dataset-dependent, and on saturated tasks a single pass already suffices. **Parameter reduction:** one block applied recursively uses $1/K$ of the parameters of K independent blocks at comparable accuracy (Tables 5, 4; Fig. 1, 3), and graded sharing schemes interpolate the trade-off (Table 6). The remaining tables probe where computation is spent (Table 7), key/value reuse (Table 8), warm-starting (Table 9) and routing configurations (Table 10); the same design decisions transfer to the pathway multi-omics cohorts (Table 11).

How few tokens suffice. Token reduction. SMART keeps only M interpretable tokens, so attention costs $O(M^2)$ rather than $O(N^2)$ in the number of genes. Accuracy is reported as M shrinks; a few dozen to a few hundred tokens recover most of the full-gene signal..

Routing and marker-token selection. Two questions about the selective recursion. Expert- vs token-choice routing allocate depth differently; and learning which genes are markers (the cross-attention router) is compared against fixed random and variance-ranked panels..

Adaptive recursion depth. The adaptive-loop result. A single pass ($K=1$), fixed-depth recursion, and adaptive per-token routed depth are compared; the mean token depth and compute saved quantify the funnel – uninformative tokens exit early while accuracy tracks fixed depth..

Recursion versus independent layers across model sizes. Recursion vs independent depth across model sizes: independent layers (Vanilla), one weight-shared block applied K times (Recursive), and adaptive routing (SMART). Recursive matches independent at a fraction of the parameters..

Parameter reduction from weight sharing. The parameter-reduction result. Applying one block recursively uses $1/K$ of the parameters of K independent blocks of the same width (a $4\times$ reduction at $K=4$)..

Weight-sharing schemes. Between fully shared and fully independent lie graded sharing schemes – Cycle, Sequence, Middle-Cycle and Middle-Sequence – trading parameters for capacity..

Where computation is spent. An interpretability view of the adaptive loop: the mean recursion depth assigned to each marker token and the fraction of tokens still active at each recursion step..

Key/value reuse across recursions. Whether the first recursion’s attention keys/values can be reused across later steps (a cache) instead of recomputed, trading a little accuracy for compute..

Warm-starting recursion from a fixed-depth model. Whether a recursive model benefits from initialising its shared block from a trained fixed-depth model and continuing training, versus training from scratch..

Routing configurations. How the routing behaves under different router heads (linear vs MLP), temperatures and load-balancing strengths, for both expert- and token-choice routing..

Transfer to pathway-informed multi-omics. Whether the same design decisions hold beyond single cells: the architecture ablation rerun on Reactome/P-NET mutation/CNV/expression cohorts..

5 Discussion

The three ingredients transfer unevenly. Parameter reduction is essentially free here: weight sharing gives an exact $K\times$ reduction at accuracy comparable to untied layers. Adaptive depth helps when the task has slack to allocate; on saturated datasets fixed depth or a single pass is already enough, and on the weakest dataset (Segerstolpe, whose genomap features are under-resolved) no architecture choice rescues it – we report these cases rather than hide them. Curated biological priors give small, mostly within-noise gains, consistent with their being warm-start regularisers rather than accuracy drivers.

Table 1: Macro-F1 as the number of marker/pathway tokens M is reduced (token reduction).

Dataset / tokens M	$M=16$	$M=32$	$M=64$	$M=128$	$M=256$
tabula_muris	36.3	53.2	65.2	70.9	76.8
pancreas	36.1	50.3	54.0	59.0	57.2
common_class	14.7	30.9	33.9	64.8	—
prototype	63.2	—	—	—	—
baron	—	—	—	—	—
segerstolpe	—	—	—	—	—

Table 2: Expert- vs token-choice recursion routing and learned vs random/variance marker panels, macro-F1.

Variant (macro-F1)	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe
expert-choice MoR (shared)	72.7 $\pm$ 1.1	53.5 $\pm$ 1.1	66.2 $\pm$ 2.9	94.2 $\pm$ 0.4	64.5 $\pm$ 0.0	5.9 $\pm$ 0.0
token-choice MoR	75.1 $\pm$ 2.1	58.7 $\pm$ 5.3	67.4 $\pm$ 3.4	94.3 $\pm$ 0.6	58.6 $\pm$ 0.0	5.5 $\pm$ 0.0
fixed-depth (no routing)	77.1 $\pm$ 2.1	53.6 $\pm$ 1.6	68.5 $\pm$ 2.7	94.1 $\pm$ 0.3	65.8 $\pm$ 0.0	9.8 $\pm$ 0.0
K=1 (no recursion)	73.1 $\pm$ 2.4	51.7 $\pm$ 5.2	66.2 $\pm$ 3.7	93.2 $\pm$ 0.5	57.5 $\pm$ 0.0	7.7 $\pm$ 0.0
independent blocks	75.5 $\pm$ 0.8	57.3 $\pm$ 1.9	66.6 $\pm$ 4.4	94.2 $\pm$ 0.4	58.3 $\pm$ 0.0	7.0 $\pm$ 0.0
random markers	74.6 $\pm$ 3.1	59.0 $\pm$ 2.0	63.6 $\pm$ 3.9	96.8 $\pm$ 0.0	65.5 $\pm$ 0.0	5.5 $\pm$ 0.0
variance markers	75.6 $\pm$ 0.7	59.2 $\pm$ 2.3	72.2 $\pm$ 1.8	95.4 $\pm$ 0.6	73.3 $\pm$ 0.0	9.6 $\pm$ 0.0

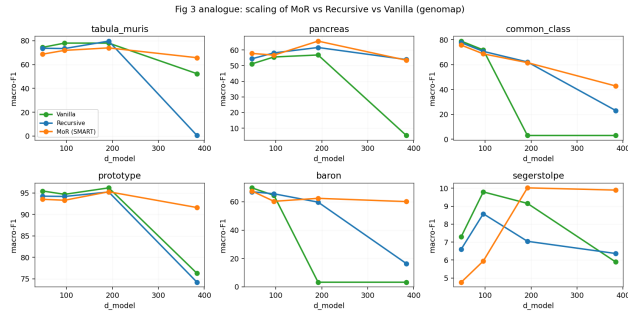


Figure 1: Macro-F1 versus model size for independent (Vanilla), weight-shared (Recursive) and adaptively-routed (SMART) stacks on each dataset.

6 Conclusion

SMART makes parameter efficiency architectural and computation adaptive for genomic classification, with interpretable marker- or pathway-token inputs. A controlled study on the genomap suite and Reactome/P-NET multi-omics validates token reduction and parameter reduction unconditionally, and adaptive recursion where the task allows – with negative results reported transparently.

References

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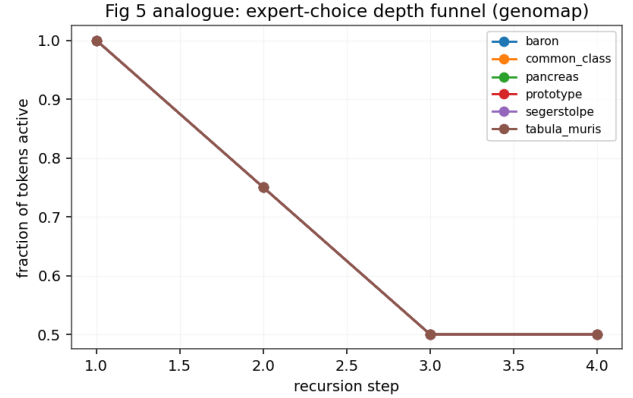


Figure 2: The expert-choice funnel: fraction of marker tokens still active at each recursion step.

Theodoris, C.; et al. 2023. Transfer learning enables predictions in network biology (Geneformer). *Nature* 618:616–624.

Table 3: Single pass ($K=1$) vs fixed-depth vs adaptive per-token routed depth, with mean depth and compute saved.

Dataset	K=1 (no recursion)	fixed depth K	adaptive depth (MoR)	mean token depth	compute saved
tabula_muris	73.1	77.1	72.7	2.75/4	31%
pancreas	51.7	53.6	53.5	2.75/4	31%
common_class	66.2	68.5	66.2	2.75/4	31%
prototype	93.2	94.1	94.2	2.75/4	31%
baron	57.5	65.8	64.5	2.75/4	31%
segerstolpe	7.7	9.8	5.9	2.75/4	31%

Table 4: Macro-F1 of independent layers (Vanilla), one weight-shared block (Recursive), and adaptive routing (SMART).

Arch x size (macro_f1)	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe
Vanilla (independent) d=48	74.5	51.1	78.8	95.5	69.8	7.3
Vanilla (independent) d=96	78.0	55.5	71.7	94.7	64.5	9.8
Vanilla (independent) d=192	78.0	56.8	3.0	96.2	3.3	9.2
Vanilla (independent) d=384	52.4	5.3	3.0	76.3	3.3	5.9
Recursive (shared) d=48	73.7	54.3	77.6	94.3	67.0	6.6
Recursive (shared) d=96	73.5	58.1	70.6	94.2	65.6	8.6
Recursive (shared) d=192	79.7	61.5	62.1	95.2	59.7	7.0
Recursive (shared) d=384	0.5	53.8	23.0	74.2	16.4	6.4
MoR (SMART) d=48	68.7	57.8	75.6	93.5	67.6	4.8
MoR (SMART) d=96	71.9	56.7	68.6	93.3	60.3	5.9
MoR (SMART) d=192	74.0	65.6	61.5	95.3	62.4	10.0
MoR (SMART) d=384	65.8	53.4	42.8	91.6	60.1	9.9

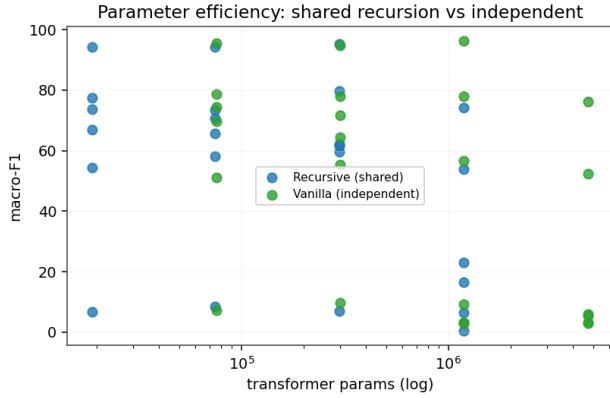


Figure 3: Accuracy versus transformer parameters for weight-shared recursion against independent layers.

Table 5: Transformer parameters of each size variant and the shared-vs-independent reduction factor.

Size	Recursions	MoR params	Vanilla params	param reduction
d_model=48	K=4	19,152	75,840	4.0×
d_model=96	K=4	75,168	299,136	4.0×
d_model=192	K=4	297,792	1,188,096	4.0×
d_model=384	K=4	1,185,408	4,735,488	4.0×

Table 6: Cycle, Sequence, Middle-Cycle and Middle-Sequence sharing between the fully-shared and fully-independent extremes.

Sharing scheme (K=6)	params	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe
shared (1 block)	74,784	75.1	61.0	68.9	93.6	60.2	3.4
Cycle (3)	224,352	80.5	54.4	71.6	95.2	70.5	6.9
Sequence (3)	224,352	78.5	57.6	0.8	93.9	66.1	4.5
Middle-Cycle (3)	224,352	80.1	56.4	67.0	94.9	61.0	7.8
Middle-Sequence (3)	224,352	80.1	56.4	67.0	94.9	61.0	7.8
independent (6)	448,704	79.9	57.0	3.0	95.4	65.3	7.2

Table 7: Mean recursion depth per marker token and the fraction of tokens still active at each step.

Dataset	mean token depth	active fraction per step (1..K)
tabula_muris	2.75/4	1.00, 0.75, 0.50, 0.50
pancreas	2.75/4	1.00, 0.75, 0.50, 0.50
common_class	2.75/4	1.00, 0.75, 0.50, 0.50
prototype	2.75/4	1.00, 0.75, 0.50, 0.50
baron	2.75/4	1.00, 0.75, 0.50, 0.50
segerstolpe	2.75/4	1.00, 0.75, 0.50, 0.50

Table 8: Recomputing vs reusing the first-step attention keys/values across recursion steps.

KV strategy (macro-F1)	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe
recompute K/V (no cache)	79.9	54.1	70.4	93.9	65.6	8.6
reuse step-1 K/V (step-cache)	71.6	58.5	71.8	93.2	59.6	6.4

Table 9: Initialising the shared block from a trained fixed-depth model vs training from scratch.

Warm-start (macro-F1)	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe
fixed-depth source	75.2	51.8	69.0	92.9	68.1	8.3
MoR from scratch	71.7	62.8	65.8	95.5	61.5	5.4
MoR warm-started	65.8	61.6	64.8	93.3	55.8	6.9
warm-start gain	-5.9	-1.2	-0.9	-2.2	-5.6	+1.5

Table 10: Router head, temperature and load balancing for the expert- and token-choice routers.

Routing config (macro-F1)	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe
expert linear	71.9	56.7	68.6	93.3	60.3	5.9
expert MLP	67.2	51.9	70.2	93.2	49.4	7.1
expert temp=2	73.5	57.6	66.7	94.1	66.2	6.5
token linear	74.7	53.7	70.1	94.5	52.6	5.4
token MLP	78.8	59.7	64.4	94.0	60.9	5.5
token balance=0.01	75.6	57.6	67.5	93.7	58.6	5.0

Table 11: The same architecture ablation on the Reactome/P-NET cohorts (mutation/CNV/expression).

Pathway cohort (macro-F1)	prostate	blca	stad	brca	panmeta_response	panmeta_subtype
adaptive MoR (shared)	50.9	43.3	38.8	27.7	93.0	50.3
independent	47.5	40.4	55.3	28.8	91.8	34.2
token-choice	80.9	43.3	37.6	30.7	91.8	57.9
fixed-depth	75.5	43.3	36.4	22.6	91.5	68.4
K=1	47.7	40.0	38.8	13.5	90.7	55.4